MIL-D-17013C(SHIPS)
9 August 1967
SUPERSEDING
MIL-D-17013B(SHIPS)
7 April 1954
(See 6.3 and 6.4)

MILITARY SPECIFICATION

DOUGH MIXING MACHINE, BAKER'S

1. SCOPE

1.1 Scope. This specification covers a two-speed, horizontal-type machine for mixing bread dough.

1.2 Classification. Electrically operated dough mixing machines shall be of the following sizes, as specified (see 6.1).

Size I - 150 pounds, finished dough capacity.
Size II - 300 pounds, finished dough capacity.
Size III - 600 pounds, finished dough capacity.

2. APPLICABLE DOCUMENTS

2.1 Not applicable to this specification.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
Standard 45 - Recommended Practice for Electrical Installation on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, N. Y. 10017.)

FSC 7320

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BAKING INDUSTRY SANITATION STANDARDS COMMITTEE (BISSC)
Sanitation Standards Number 20.

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 511 Fifth Avenue, New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning production, a sample shall be tested as specified in 4.3 through 4.3.3 (see 6.2).

3.2 Material. Materials used shall be equal to or better than the quality used by the manufacturer in his regular commercial production. Stainless steel where required, shall be AISI types 302 or 304. Materials subject to contact with the product shall be of composition and finish at least equal to or better than required by BISSC Standard No. 20.

3.3 Design and construction. The machines shall be compact and self-contained, and shall consist of a frame supporting a mixing bowl, in which a rotating agitator is driven from a motor by means of V-belts, silent or roller chain, gearing or a combination of such driving means. The machine shall be capable of thoroughly developing bread dough from the ingredients listed in 4.3.1, and in quantities for specified size (see 4.3.1.1). The machine shall thoroughly mix dough without slicing or other outside assistance, and shall leave no dry flour in the tank. The machine shall be of rigid construction, free from objectionable vibration, and quiet in operation.

3.3.1 Frame. The frame shall be of steel and suitable for supporting the mixing bowl when operated at full load. The base of the frame shall be a sanitary wash-under type equipped with four steel legs, not less than six inches high, drilled for four 5/8-inch minimum diameter hold-down bolts, one at each corner.

3.3.2 Mixing bowl. The mixing bowl (bowl sheet, bowl ends, and bowl cover) shall be constructed of stainless steel sheets or plates. The bowl sheet and cover shall be not less than 0.225 inch thick. The bowl ends shall not be less than 0.600 inch thick. A small fillet of stainless steel shall be continuously welded, ground smooth, and coved to a minimum radius of 0.125 inch around the entire juncture of bowl sheet and bowl ends. Trunnions shall be made of steel or ductile iron and shall be welded, bolted or riveted to both bowl ends. When tilted to the maximum position, the bowl shall permit emptying the dough into a trough 26 inches high.

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The mixing bowl cover shall be provided with a flour inlet gate not less then 12 inches in diameter and an adequate water inlet. The bowl shall be fitted with an indicating thermometer, adequately protected, at a convenient height on the front of the bowl for observation. The thermometer shall indicate the temperature of the dough within the bowl.

3.3.3 Agitating device. The agitating device shall be of the single arm nonreversing type with shafts extending through the ends of the mixing bowl. Agitating device and shafts shall be made of stainless steel. Either one or both shafts shall be fitted on the outer end with a driving means (such as a gear or sprocket) for connection to motive power. The agitator shall have two speeds, operating in the range of 20 to 22 revolutions per minute in low speed and in the range of 40 to 44 revolutions per minute in high speed, obtained by means of a two-speed electric motor.

3.3.4 Time clocks. Two time clocks with splashproof faces shall be mounted on the front of each machine. The clocks shall be so connected as to automatically switch the mixer from low speed to high speed when the predetermined low speed time has elapsed, and to automatically shut the mixer off when the full predetermined mixing time (low speed time plus high speed time) has elapsed.

3.3.5 Safety switch. All machines shall be fitted with a safety switch controlled by a JOG button so that both hands of the operator must simultaneously press the JOG button and the START button to rotate the agitator when the bowl is tilted 6 inches or more from the vertical or operating position. The safety JOG button and the START button shall be simultaneously available to the operator but located not less than 10 inches apart.

3.3.6 Dumping device. Devices for dumping or emptying the bowl shall be provided as follows:

3.3.6.1 Sizes I and II machines. A manually operated device, incorporating a worm gear set, arranged so that the bowl is at all times automatically locked in any given position.

3.3.6.2 Size III machines. A power dump, complete with adequately powered reversible brake type motor, controls and push button stations labeled UP, DOWN and STOP, together with a safety switch, including a safety JOG button to prevent the bowl from being closed beyond four-fifths of the total opening unless the operator maintains a two-hand contact to cause the tilting motor to complete the bowl closure.

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3.3.7 Hood. A hood made of stainless steel shall cover the top of the machine. The hood shall be supported by the frame of the machine and the front portion shall be hinged.

3.3.8 Housing. Each machine shall be completely enclosed and fitted with safety guards so that moving parts (gears, chains, etc.) do not present a safety hazard. Safety guards shall be readily removable and readily removable access doors shall be provided, as necessary, in the housing for cleaning and maintaining the machine.

3.3.9 Shafts. Shafts, except agitating device shafts, shall be made of high grade, cold-rolled or cold-drawn machinery steel, and shall run in ball bearings or bearings having bronze or other bushings satisfactory to the activity concerned.

3.3.10 Lubrication. Pressure gun lubrication fittings shall be provided at readily accessible points for all bearings requiring grease lubrication. Gear boxes, where used, shall be provided with accessible screw plugs for filling and draining. Lubricants of all types shall be prevented from entering the mixing bowl.

3.3.11 Agitator shaft seals. Seals shall be provided at each end of the bowl and shall be of a design that will prevent any leakage of the dough or ingredients at these points. Seals shall be independent of the agitator shaft bearings and shall not carry any part of the agitator load. No absorbent packing material shall be used. Seals shall be accessible and removable for cleaning and renewal of the seal rings.

3.3.12 Drives.

3.3.12.1 Gear or chain drive. All pinions and gears shall be of steel and have cut teeth, and be designed for quiet operation and long life. The cutting of the gears, alignment of the shafts and gears, and deflection of these shafts shall be such that the error in action shall not exceed 0.002 inch. Spiral bevel gears shall be substantially mounted to reduce the deflection and endwise displacement of the gears under load. Spiral bevel gears shall be mounted on either ball bearings or taper roller bearings to fully provide for both thrust and radial loads. Means shall be provided for accurately positioning the spiral bevel gears on their shafts. All gears shall be fitted to their shafts or bearings. Where the gears transmit torque through the shafts, the gears shall be keyed to the shaft and means shall be provided to prevent axial movements along the shaft. The use of set screws for holding gears on shafts is not acceptable. Where nuts are used for securing gears, and unless they are so threaded that they will tighten whenever the machine starts, they shall have locknuts. The use of lock washers without locknuts is not acceptable. All keys and keyways shall conform to the current standards of the

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American Standards Association in regard to dimensions and tolerances for corresponding shaft diameters. The fit of the gears on their shafts or bearings shall be American Standards Association Class 3 fit, or better. All gears, sprockets, and chains shall be adequately lubricated and enclosed in chambers sealed against loss of lubricant. Gears and chains shall be readily accessible for inspection. Studs and shafts for the mounting of bearings are not acceptable unless the elongation in a 2-inch sample of the metal is at least 10 percent. Means shall be provided for adjustment of chain tension.

3.3.12.2 V-belt.

3.3.12.2.1 Belts, when used, shall be V-type provided in accordance with the belt manufacturer's recommendations based on load ratings, with adequate safety factors, greater than the maximum belt loads resulting under extreme operating conditions. The V-belts shall have a minimum service life of 5000 hours of machine operation.

3.3.12.2.2 The diameter of pulleys shall not be less than recommended by the belt manufacturer.

3.3.13 Electrical requirements. Unless otherwise specified in the contract or order (see 6.1), all electrical equipment shall conform to the recommended practice for electrical installation aboard ship as specified by the requirements in IEEE Standard 45. All electrical equipment shall be designed for operation on an ungrounded electrical system but shall operate satisfactorily and without hazard to personnel or equipment with an accidental ground on any conductor. The manufacturer shall provide transformers, electrical relays, rectifiers or converters, as necessary, for other required electrical characteristics within the machine that differ from specified source electrical characteristics.

3.3.13.1 Motors. Motors shall be suitable for the required application under all operating conditions without exceeding permissible temperature rise. In addition, main drive motors shall have the following characteristics:

Ambient temperature	50°C.
Service	C.
Enclosure	Open and dripproof.
Cooling	Natural or fan.
Horsepower	To suit load.
Speed	Two-speed, constant for both windings.
Duty	Continuous.
Voltage	As specified (see 6.1).
Insulation	Class B, or better.
Bearings	Ball or sleeve.

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3.3.13.2 Controllers. Contactors shall be used for all power circuits and shall switch all three lines of three phase circuits and both sides of single phase circuits. All control circuits shall operate at 115 volts or less. Solenoid valves, if used, shall have epoxy encapsulated coils. The control equipment shall have the following characteristics:

Ambient temperature	50°C.
Enclosure	Dripproof.
Operation	Magnetic.
Type	AC: Across-line. DC: Resistor.
Function	Motor starting, inching, speed changing.
Protection	Thermal overload and low voltage protection.
Duty	To suit application.

3.3.13.3 Wiring. Wiring attached to the machine shall be installed in electrical metallic tubing and be suitably connected. Wiring on the machine to be connected to devices beyond the machine, such as power disconnect switch, shall terminate in an outlet box or boxes for conduit connection to the devices beyond the machine.

3.4 Capacity and height. The respective capacity and height shall be as specified in table I.

Table I. Capacity and height.

SIZE	CAPACITY OF BOWL (MINIMUM) POUNDS OF FINISHED BREAD DOUGH	HEIGHT OVERALL (MAX.) INCHES
I	150	65
II	300	65
III	600	70

3.4.1 Total bowl capacity. The bowl size shall be of sufficient capacity to prevent the specified ingredients for finished dough weights from splashing or spilling out while being mixed at the high speed.

3.5 Finish. The machines shall be finished in accordance with the manufacturer's standard practice.

3.6 Identification plate. An identification plate of durable material shall be secured in a conspicuous location on each machine. Marking shall be permanent and legible. The plate shall include information relative to the name of machine, specification size and capacity, manufacturer's name, type,

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serial number, date of manufacture, contract number, electrical characteristics, and the applicable Federal stock number (when indicated in contract or order).

3.7 Repair parts. Repair parts shall accurately fit the machine for which they are furnished, and shall be interchangeable with machines of the same type and manufacture. Repair parts shall be furnished, in general, as follows, or as determined upon completion of Form 4786 (Copies of form 4786 may be obtained from the cognizant Naval Inspector):

3.7.1 Mechanical. Mechanical onboard repair parts shall be furnished with each machine as follows:

1 set of packing or agitator shaft seals.
1 set of motor bearings for each size and type of motor.
1 gear of each type and size used in drive mechanisms.
1 each type and size drive belt or chain.
1 each type and size of bearing other than motor.

3.7.2 Electrical. Electrical onboard repair parts shall consist of one set each, for each size component used, of those items subject to normal wear or breakdown, such as motor brushes; starter coils; solenoid coils; and controller parts subject to wear, such as line contactors.

3.8 Technical manuals. Technical manuals shall be of a commercial type, consisting, in general, of the manufacturer's standard literature, illustrations or drawings to concisely outline the proper installation, operation, lubrication, maintenance, and parts identification of the machine. Each manual shall be suitably bound in a self-cover of durable quality. Two copies of the manual shall be submitted to the ordering activity for approval and assignment of a NAVSHIPS number. Two copies of the approved manual shall be packed with each dough mixing machine. Additional copies of the approved manual for stock shall be as specified (see 6.1).

3.9 Workmanship. The design, materials of construction, fabrication, finish and operation of each complete dough mixing machine shall be suitable for the purpose intended and be free of all flaws that may affect serviceability and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General. Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or material shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection and acceptance for the Government will be made after completion of the preproduction tests. Final acceptance inspection shall determine compliance with all requirements of this specification and shall include the tests under 4.3, 4.3.1, 4.3.2 and 4.3.3.

4.2 Preproduction inspection. Prior to the start of production and within 45 days after the date of award of the contract or order, one complete dough mixing machine of each size under contract or order shall be subjected to the tests under 4.3, 4.3.1, 4.3.2,and 4.3.3. These tests shall be witnessed by an authorized Government representative and production of the remaining dough mixing machines under contract or order is contingent on the preproduction machine satisfactorily passing all tests.

4.3 Tests. Each preproduction dough mixing machine shall be assembled and set up in a space available to the supplier. After assembly and connection to power source, the machine shall be operated for one hour without load to determine that all parts function properly and that motors and bearings do not overheat.

4.3.1 Functional test. Each preproduction dough mixing machine shall be functionally tested by mixing the ingredients for white bread as hereinafter specified. The machine shall fully develop not less than one batch of bread dough from the quantities of ingredients specified in 4.3.1.1 without splashover of ingredients or overheating of motors and bearings. Ingredients to be used for the white bread dough shall consist of the following:

Table II - Ingredients for dough.

Ingredients	Weights	Method
Yeast, active dry. 1/ Water, warm (110°F.) 2/ Sugar	1½ lb. 7 lb. 3/4 oz. 8½ oz.	1. Sprinkle yeast on top of water. Let stand 5 minutes, then stir. Add sugar and stir until dissolved. Let stand 10 minutes, then stir again.
Water (65°F.) variable. 2/ Milk, nonfat, dry. Sugar. Dough conditioner (optional).	45 lb. 1 3/4 oz. 4 lb. 3 3/4 oz. 2 lb. 7½ oz. 7 oz.	2. Place water in mixer Add milk, sugar and dough conditioner. Mix on low speed 2 or 3 minutes to break up any lumps.

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Table II - Ingredients for dough, con't

Ingredients	Weights	Method
Flour, hard wheat, sifted. Shortening, soft.	62 lb. 1 oz. 3 lb. 14 oz.	3. Add flour and start mixer on low speed; add shortening and yeast suspension and mix until flour is thoroughly moistened. 4. Mix on high speed 10 minutes. 5. Let rise in mixer bowl 20 minutes.
Flour, hard wheat, sifted. Salt. Cornstarch.	20 lb. 7 oz. 1½ lb. 14 oz.	6. Mix and sift flour, salt and cornstarch; add with mixer on slow speed (2 min.). Mix on high speed 10 minutes or until dough is smooth and elastic.

1/ 1 lb. 1 oz. of compressed yeast may be substituted for active dry yeast.
2/ Water temperature and fermentation are based on a final dough temperature of 90°F.

4.3.1.1 The following quantities shall be used for the different size machines:

Size I - The recipe quantities.
Size II - Double the recipe quantities.
Size III - Four times the recipe quantities.

4.3.2 Power input. The power input shall be measured during the functional test under 4.3.1.

4.3.3 Insulation. The insulation resistance of power carrying electrical components shall be measured with a 500 volt, d.c., insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

4.4 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to assure compliance with section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.

5.1.1 Machine, dough mixing, and attached parts.

5.1.1.1 Preservation and packaging. Each dough mixing machine and detached parts shall be preserved and packaged in accordance with the supplier's commercial practice to provide protection from corrosion and physical damage in general handling, storage and issue.

5.1.1.2 Packing. Each dough mixing machine and detached parts shall be packed in containers in such a manner as to insure acceptance by common carrier and safe delivery at destination. Containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include item name; Federal stock number (when provided); contract or order number; contractor's name; and destination.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Electrical standards, if other than specified (see 3.3.13).
(d) Source electricity (see 3.3.13.1).
(e) Quantity of manuals for stock (see 3.8).
(f) Preservation, packaging, packing or marking requirements, if other than specified (see 5.1).

6.2 Preproduction.

6.2.1 Invitation for bids should provide that the Government reserves the right to waive the requirements for preproduction samples to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is currently appropriate for the pending procurement.

6.3 Sizes of dough mixing machines covered in this specification replaced those covered in MIL-D-17013B(SHIPS) as follows:

This specification		MIL-D-17013B(SHIPS)
Size I	replaces	Size C
Size II	replaces	Size A
Size III	replaces	Size B

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6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated.. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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